

The Ringdown Register

CURRICULUM VITÆ OF ANUJ KANKANI · GRAVITATIONAL-WAVE PHYSICS

Gravitational Waveform Modeling — Numerical Relativity — General Relativistic Particle-in-Cell Simulation

VOL. I — No. 1

MORGANTOWN, WEST VIRGINIA — SUNDAY, AUGUST 16, 2026

PRICE FIVE CENTS

WEST VIRGINIA UNIVERSITY · DEPARTMENT OF PHYSICS & ASTRONOMY

KANKANI MODELS BLACK-HOLE MERGERS AND THE PLASMA THAT SURROUNDS THEM

SEVEN PUBLICATIONS, SIX OF THEM FIRST-AUTHORED; THREE
REFEREED AND IN PRINT, FOUR SUBMITTED AND UNDER REVIEW

PRIMARY DEVELOPER OF gwBOB — CONTRIBUTOR TO THE EINSTEIN TOOLKIT, TO ENTITY
AND TO NRPy — CO-DEVELOPER OF LISAWAVE-JAX

Doctoral Candidate at West Virginia University — Expected 2027 — Advisor, Sean T. McWilliams

ANUJ KANKANI is a doctoral candidate in the Department of Physics & Astronomy at West Virginia University, Morgantown, under Sean T. McWilliams, expecting to graduate in 2027.

He works on gravitational waveform modeling, numerical relativity, and General Relativistic Particle-in-Cell (GRPIC) simulations. He took his BS in Physics (Honors) at Texas A&M University in May 2021, with

minors in Computer Science and Astrophysics.

Of his seven publications, three are refereed and in print — two in *Physical Review D*, one in *Physical Review E* — and are set out at the foot of this page. Four more are submitted and under review; they are announced separately, on Page Two. Six of the seven carry his name first.

He is the primary developer of gwBOB, an open-source Python pack-

age implementing the Backwards One Body waveform model, contributes to the Einstein Toolkit, to Entity and to NRPy, and co-develops LisaWave-JAX. All five are on Page Three.

He is a member of the LISA Consortium, in its waveform and astrophysics working group. His engagements are on Page Two; his service and outreach, on Page Three.

THIS OFFICE

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TERMS OF STUDY

WEST VIRGINIA UNIVERSITY

PhD in Physics — in progress
Expected graduation 2027. Advisor:
Sean T. McWilliams. Grade-point
average 3.9 / 4.0.

TEXAS A&M UNIVERSITY

BS in Physics (Honors)
College Station, Texas. August
2017 – May 2021. Minors in Com-
puter Science and Astrophysics.
Grade-point average 3.84 / 4.0.

HONOURS CONFERRED

RUBY DISTINGUISHED DOCTORAL FELLOW 2021– 2024

Awarded to outstanding doctoral
students in STEM fields at West
Virginia University.

NASA WV SPACE GRANT GRADUATE FELLOWSHIP 2024–2025

For the proposal “Modeling Black
Hole Mergers with a Dynamical
Background Spacetime.”

MOHINDAR SEEHRA RESEARCH AWARD 2025

Departmental award recognizing
outstanding research efforts in
Physics.

PUBLISHED AND REFEREED

Three papers, refereed and in print. Submitted work is announced separately, on Page Two.

PHYSICAL REVIEW D ·
2025

MODELING RELATIVE PEAK TIMES OF GRAVITATIONAL WAVE HARMONICS

PHYS. REV. D 112, 124051
(2025)

Anuj Kankani and Sean T.
McWilliams

ARXIV:2506.17492

**

First author · Refereed

PHYSICAL REVIEW D ·
2024

TESTING THE BOUNDARY-TO- BOUND CORRESPONDENCE WITH NUMERICAL RELATIVITY

PHYS. REV. D 110, 064033
(2024)

Anuj Kankani and Sean T.
McWilliams

ARXIV:2404.03607

**

First author · Refereed

PHYSICAL REVIEW E ·
2021

OPTIMIZATION AND BENCHMARKING OF THE THERMAL CYCLING ALGORITHM

PHYS. REV. E 104(3), 035302
(2021)

A. Barzegar, A. Kankani, S. Mandrà
and H. G. Katzgraber

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Second author · Refereed

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ANNOUNCEMENTS

NOTICE TO THE READER. *The four papers below have been SUBMITTED to the journals named and have NOT completed refereeing. They are printed here as announcements, in a lighter face, and are deliberately set apart from the three refereed papers on Page One.*

SUBMITTED — UNDER REVIEW

THE DIRECT WAVE IS NOT A
MEANINGFUL TEST OF HORIZON
PROPERTIES

SUBMITTED TO *Physical Review Letters*

Anuj Kankani and Sean T. McWilliams

ARXIV:2607.02380 · FIRST AUTHOR

SUBMITTED — UNDER REVIEW

DIRECT WAVES IN BLACK-HOLE
BINARY MERGERS: INSIGHTS FROM
THE BACKWARDS ONE BODY MODEL

SUBMITTED TO *Physical Review D*

Anuj Kankani and Sean T. McWilliams

ARXIV:2603.15474 · FIRST AUTHOR

SUBMITTED — UNDER REVIEW

gwBOB: A PYTHON PACKAGE FOR
ANALYTICAL MERGER-RINGDOWN
GRAVITATIONAL WAVEFORMS

SUBMITTED TO *Journal of Open Source Software*

Anuj Kankani, Angel Morales, Suchindram Dasgupta and Sean T.
McWilliams

FIRST AUTHOR

SUBMITTED — UNDER REVIEW

BOB THE (WAVEFORM) BUILDER:
OPTIMIZING ANALYTICAL BLACK-
HOLE BINARY MERGER WAVEFORMS

SUBMITTED TO *Physical Review D*

Anuj Kankani and Sean T. McWilliams

ARXIV:2510.25012 · FIRST AUTHOR

ENGAGEMENTS AND APPEARANCES

Twelve meetings and workshops — nine talks, two posters, one attended.

2026 PLASMAS AROUND BLACK HOLES AND NEUTRON STARS (SCEECs)
POSTER
“Driving Temperature Anisotropies in the Jets of Black Holes”

2026 16TH INTERNATIONAL LISA SYMPOSIUM
CONTRIBUTED TALK
“BOB the (Waveform) Builder + Direct Waves”

2026 APS GLOBAL PHYSICS SUMMIT
CONTRIBUTED TALK
“BOB the (Waveform) Builder: Optimizing Analytical Black-Hole Binary Merger Waveforms”

— **CENTER FOR GRAVITATIONAL WAVES AND COSMOLOGY SYMPOSIUM**
CONTRIBUTED TALK · YEAR NOT STATED
“Modeling Extreme Gravity: From Relativistic Plasmas to Gravitational Waves”

2025 PROSPECTS IN THEORETICAL PHYSICS — GRAVITATIONAL WAVES FROM THEORY TO OBSERVATION
ATTENDED

2025 APS GLOBAL PHYSICS SUMMIT
CONTRIBUTED TALK
“Merger-Ringdown Modeling of Gravitational Waves Beyond the Dominant Mode”

2025 AAS WINTER MEETING
CONTRIBUTED TALK
“Understanding the Origins of Black Hole Superkicks”

2024 NORTH AMERICAN EINSTEIN TOOLKIT WORKSHOP
THIRTY-MINUTE TALK
“Understanding the Origins of Black Hole Superkicks”

2024 PENN STATE NEIGHBORHOOD WORKSHOP
CONTRIBUTED TALK
“Understanding the Origins of Black Hole Superkicks”

2024 APS APRIL MEETING
CONTRIBUTED TALK
“Testing the Boundary to Bound States Dictionary with Numerical Relativity”

2023 AAS WINTER MEETING
POSTER
“Effect of Baryons on the Dynamical-Friction Timescale of Merging Dark Matter Halos”

2022 APS APRIL MEETING
CONTRIBUTED TALK
“Effect of Baryons on the Dynamical-Friction Timescale of Merging Dark Matter Halos”

SOFTWARE AND CODE

*Five open-source projects — primary developer of one, co-developer of another, contributor to three.***gwBOB****PRIMARY DEVELOPER***github.com/AnujKankani/BackwardsOneBody*

Open-source, user-friendly Python package implementing the Backwards One Body waveform model, with tools for analysis and NR comparison.

LisaWave-JAX**CO-DEVELOPER — IN DEVELOPMENT**

JAX-based LISA data-analysis tools for massive black holes and galactic binaries.

Entity**CONTRIBUTOR***entity-toolkit.github.io/wiki*

Kokkos-based General Relativistic Particle-in-Cell code for plasma astrophysics.

Einstein Toolkit**CONTRIBUTOR***einsteintoolkit.org*

Open-source high-performance-computing codebase for numerical relativity.

NRPy**CONTRIBUTOR***github.com/nrpy/nrpy*

Python-based symbolic code-generation toolkit for relativistic astrophysics.

CIVIC NOTICES

*Consortium membership, departmental service, outreach and community work.***LISA CONSORTIUM**

Member of the waveform and astrophysics working group.

ASTRONOMY JOURNAL CLUB — ORGANIZER

Organized and led the department's weekly journal club, facilitating discussions on recent research in astronomy and astrophysics.

COMMUNITY SERVICE — 200 HOURS

Two hundred hours of community service completed as a graduate student, focused on environmental restoration and STEM outreach.

PHYSICS AND ASTRONOMY GRADUATE STUDENT ORGANIZATION

Regularly involved in STEM outreach such as Physics Demo Day and the pumpkin drop contest.

PHYSICS OUTREACH

Talks to middle- and high-school groups about black holes and gravitational waves; regular volunteer at physics outreach events such as Physics Demo Day; writes articles on astrophysics for local organizations.

TERMS OF CORRESPONDENCE

BY POST

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Morgantown, West Virginia

BY WIRE

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COLOPHON AND STATEMENT OF THE PRESS

The Ringdown Register is not a newspaper. It is the curriculum vitæ of Anuj Kankani, set in the manner of a broadsheet front page of the eighteen-nineties. The title is invented; no publication of that name exists, and none is imitated. Every fact printed in these columns is drawn from that curriculum vitæ, and nothing has been added for effect — there are no quotations, no correspondents, no reports, and no

assessment of the work beyond what its author has stated. The volume number, the price line and the ornaments are costume; the dateline is the day of setting. Set in Old English Text MT for the nameplate, Bodoni MT for headlines, Engravers MT for standing heads, and Times New Roman for text and decks. Composed in HTML and printed to PDF. Morgantown, West Virginia, August 2026.